

The L1 figure uses Experiment 1’s whole-tensor canonical layouts, stratified over all score components before profiling. The H100 L1 comparison uses the issue quotient Q_{32B} , matching the counter’s 32-byte sector unit.¹ It gives perfect rank agreement for all six kernels with layout variation; correlations with downstream L2 activity are less uniform.

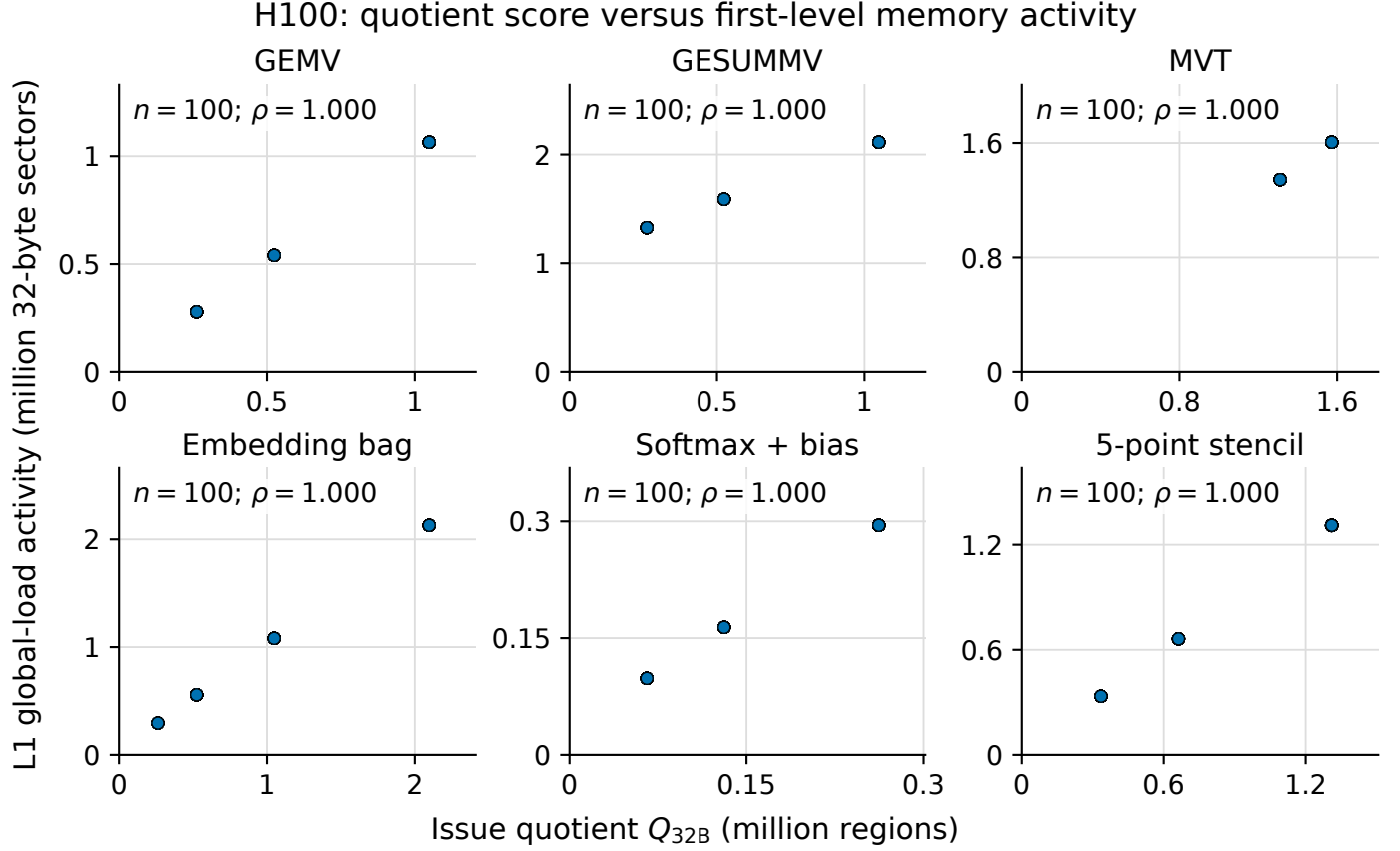


Figure 1: Issue quotient versus first-level memory activity for all six pilot kernels on NVIDIA H100. The horizontal axis is the target operand’s dynamic issue quotient Q_{32B} (component `issue.g32.stream.load.32B`); the vertical axis is whole-kernel global-load L1TEX sectors, measured by `l1tex__t_sectors_pipe_lsu_mem_global_op_ld.sum`. Both axes use millions and start at zero; panel scales differ. Each point is one distinct whole-tensor canonical mapping from the all-scope-stratified Experiment 1 panel. Repeated coordinates overlap; n counts mappings, not visible markers. Points are medians of three independent profiler-launch medians, each over 20 steady-state dispatches under the warm-cache protocol. All three launch medians coincide for every mapping, so error bars have zero height. Spearman’s ρ uses average ranks for ties. The quotient scores only the varied operand, whereas the counter includes all kernel loads; offsets therefore need not vanish.

¹NVIDIA Nsight Compute, Profiling Guide: metrics and memory workload analysis.

The L2 comparison summarizes the recorded pre-packet pilot results for G_C (Experiment 1) and $GL(p, 2)$ (Experiment 3), using the median correlation across six kernels.

Layout family	H100	MI300A
G_C	0.750	0.983
$GL(p, 2)$	0.785	0.967

Table 1: Median per-kernel Spearman correlations between J_{area} and L1-to-L2 read requests, using each device’s `l1_to_l2` tau profile and all-scope stratification. G_C denotes Experiment 1’s whole-tensor canonical layouts; $GL(p, 2)$ denotes Experiment 3’s invertible binary inner-tile layouts with row-major outer tiles. Each cell is the median of six per-kernel correlations (bias+ReLU excluded), each computed across 100 distinct mappings with average ranks for ties. H100 uses `lts__t_requests_srcunit_tex_op_read.sum`; MI300A uses `TCP_TCC_READ_REQ.sum`. Each counter observation is the median of three independent profiler-launch medians, each over 20 steady-state dispatches with warm caches. These are recorded pre-packet pilot results; device-specific tau weights were fitted using pilot counters.